Control Structure Requirements for Transmission System Operators
and Their Relevance to Hydropower Control and Simulation

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1. Introduction

Transmission System Operators (TSOs) such as Statnett in Norway are responsible for maintaining real-time balance between electrical generation and consumption to ensure secure and stable grid operation. The balance is maintained by procuring and activating various ancillary services including Frequency Containment Reserves (FCR), automatic Frequency Restoration Reserves (aFRR), and manual Frequency Restoration Reserves (mFRR). Hydropower plants often provide these services due to their fast and flexible response capabilities.

Control structure requirements from a TSO refer to the technical and functional specifications that generating units and their control systems must satisfy to successfully participate in real-time balancing markets and deliver the requested grid services.

2. TSO Balancing Markets and Control Signals

In the Nordic power system, the balancing markets are structured to facilitate frequency control and supply-demand balance across operational timeframes. FCR (primary reserves) and aFRR (secondary reserves) are activated automatically based on system frequency deviations, while mFRR (tertiary reserves) is activated on a slower timeframe (e.g., within 15 minutes) in response to broader system needs. The markets use bidding mechanisms where Balancing Service Providers (BSPs) offer capacity and energy bids which are activated by TSOs such as Statnett according to need and price merit order.

3. Control Structure Requirements Defined

Control structure requirements from a TSO typically include:

3.1 Automatic Reception and Execution of Set-Points

TSOs generate control set-points (e.g., active power adjustments) and transmit them to BSPs through secure real-time protocols like ICCP. The BSP's local control system must automatically receive and interpret these set-points and integrate them with its unit control logic. The set-point acts as a deviation from a baseline generation schedule, and the control system must translate this into physically realizable control actions on the generating unit.

3.2 Functional Control Architecture

The local control system architecture must:

- Implement rate limits, ramp rates, and safety constraints to ensure that physical actuators (e.g., turbine wicket gates and governor valves) respond without compromising mechanical or hydraulic stability.
- Avoid closed-loop local frequency functions that conflict with the goal of the centralized TSO control loop for aFRR delivery.
- Provide monitoring and verification that signals and set points are within contractual limits and operational constraints.

This ensures that delivered control actions align with the requested balancing service without unintended interactions between local and systemwide controllers.

3.3 Monitoring and Feedback

To demonstrate compliance and track performance, the BSP's control system must provide real-time measurements of power output, frequency, and reserve contribution. These data must have sufficient resolution (e.g., 0.01 MW and 0.01 Hz) and be logged for TSO evaluation during prequalification and real operation.

3.4 Communication Availability Requirements

Control structure requirements also include communication reliability. The interface between TSO and BSP (for example ICCP links) must meet availability thresholds (e.g., 99% uptime) and include timeout mechanisms to manage temporary signal loss. In case of communication failure, the local control must maintain the last known set point for a configurable period to avoid abrupt changes.

4. Relation to Hydropower Control and Simulation

Hydropower plants provide ancillary services by adjusting turbine output in response to TSO control signals. Incorporating TSO control structure requirements into hydropower control simulations enhances realism and enables testing under realistic operating conditions.

4.1 Parameters and Limits in Simulation

Simulations should include:

- Rate limits and ramp profiles for actuator motion (e.g., guide vane opening), reflecting actual hydraulic servomechanism constraints.
- Dynamic response functions (e.g., governor, turbine model) capable of following set-points within defined time constraints (such as Full Activation Time for aFRR, often targeted within 2–5 minutes).
- Communication delay and timeout mechanisms to emulate real TSO–BSP interactions.

These aspects ensure that your hydropower model captures realistic response behaviors, rather than idealized instantaneous control actions.

4.2 Control Structure in Real-Time Models

In real time, TSO control signals are effectively references for power output that must be translated into actuator commands. For example, an aFRR set-point can be treated as a desired power offset that a governor block seeks to achieve using rate-limited actuator commands. The control structure would include:

- Signal reception block (handling set-point messages from the TSO)
- Rate limiter/ramp function (ensuring actuator constraints)
- Governor and turbine dynamics (translating set-points into mechanical motion)
- Power output feedback (sending current generation values back to the TSO)

Incorporating this layered control structure into simulation environments like MTK.jl or Modelica allows for rigorous system testing and prequalification prior to field deployment.

4.3 Market Prices and Resource Participation

TSOs dispatch ancillary services based on both technical needs and economic principles. Balancing markets use marginal pricing: the price for activated reserves is often set by the most expensive accepted bid, and all accepted bids receive that price. BSPs must therefore strategize not only on control performance but also on participation costs and price expectations for energy and capacity bids in balancing markets.

5. Conclusion

TSO control structure requirements encompass automatic reception and execution of set-points, structured functional control hierarchy, monitoring and reporting requirements, and communication reliability. These requirements are essential for integrating hydropower plants into real-time balancing markets. For hydropower control and simulation, this means modeling not just plant dynamics but also communication interfaces, actuator limits, ramp rates, and TSO interaction protocols. Including these elements increases the fidelity of simulations and supports performance evaluation relative to actual market and control requirements.

References

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